302. Which of the following statements about the study is true?

- Cluster 1 is the only cluster with a median learning score that is greater than the overall median learning score.
- Cluster 1 is the one of the study, those 300 students were given an hour-long class on how to take notes in online form.
- Cluster 1 is the only cluster with a median learning score that is greater than the overall mean learning score.
- Cluster 1 is the cluster with the lowest median learning score and the smallest interquartile range of learning scores.

303. The statement "Students who spend more time on their homework tend to have higher test scores" is an example of:

- The expression variable is a dummy variable.
- The response variable is whether training was received or taking online notes.
- The explanatory variable is the amount of time.
- The statement is the lesson on biology.

304. Which of the following is true? Students who are good at taking notes are generally also organized.

305. Based on the bar chart, which of the following statements must be true?

- The distribution of learning scores for the group who received training and who took online notes is approximately normal.
- The distribution of learning scores for the group who received training and who did not take online notes is approximately normal.
- The distribution of learning scores for the group who received training and who took online notes is approximately uniform.
- The distribution of learning scores for the group who received training and who did not take online notes is approximately uniform.

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- Students who were given training and who took online notes tend to have higher learning scores than students who were given training and who did not take online notes.
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A local park has a trail where hawks, large predatory birds, are regularly spotted. Let random variable X represent the number of hawks spotted in a randomly selected day from the trail. The probability distribution for X, based on data collected over several years, is shown in the table.

Probability Distribution of the Number of Hawks Spotted in a Day

x	0	1	2	3	4	5
$P(X=x)$	$\frac{1}{20}$	$\frac{12}{20}$	$\frac{4}{20}$	$\frac{1}{20}$	$\frac{1}{20}$	$\frac{1}{20}$

Based on this probability distribution, the expected number of hawks spotted in a day from the trail is 1.6 hawks.

A. $\sigma_X = 0.20$

B. $\sigma_X = 1.08$

C. $\sigma_X = 1.16$

D. $\sigma_X = 1.34$

E. $\sigma_X = 1.70$

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344. Assuming the report is correct, what is the probability that at least 650 adults from the sample will indicate that they wash their bedsheets at least once per week?

According to a report, 44 percent of adults indicate that they wash their bedsheets at least once per week. A random sample of 1,000 adults will be selected.

A. $P\left(z > \frac{0.65-0.44}{\sqrt{\frac{0.8361 \cdot 0.41}{1,000}}}\right)$

B. $P\left(z > \frac{0.65-0.44}{\sqrt{\frac{0.6015 \cdot 0.65}{1,000}}}\right)$

C. $P\left(z > \frac{650-440}{\sqrt{\frac{0.5565 \cdot 0.65}{1,000}}}\right)$

D. $1 - P\left(z > \frac{0.65-0.44}{\sqrt{\frac{0.4015 \cdot 0.65}{1,000}}}\right)$

E. $1 - P\left(z > \frac{0.65-0.44}{\sqrt{\frac{0.6015 \cdot 0.65}{1,000}}}\right)$

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345. Of the following, which experimental design would be best for the veterinary student to use?

A veterinary student is conducting an experiment to see whether using a certain medication can reduce symptoms of arthritis in rabbits. Arthritis is a condition that makes it difficult for rabbits to use one or more of their limbs. The severity of the arthritis will be based on an evaluation of each rabbit by a veterinarian who does not know whether the rabbits received medicine. Twenty rabbits of different sizes, each with arthritis in one or more limbs, will be randomly assigned to receive medicine or no medicine. The veterinary student believes larger rabbits may respond better to the medicine than smaller rabbits will.

A. A matched pairs design using randomly paired rabbits

B. A matched pairs design pairing rabbits of similar size

C. A design in which the ten largest rabbits are assigned to receive the medicine and the remaining rabbits are assigned to not receive the medicine

D. A design in which the five largest rabbits and the five smallest rabbits are assigned to receive the medicine and the remaining rabbits are assigned to not receive the medicine

E. A completely randomized design in which ten rabbits are randomly assigned for each treatment

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346. If one customer from the restaurant is selected at random, which of the following is closest to the probability that the customer ordered a drink other than water and did not order dessert?

In a certain restaurant, 70 percent of the customers order a drink other than water and 44 percent of the customers order dessert. Of the customers who order a drink other than water, 58 percent order dessert.

A. 0.034

B. 0.266

C. 0.294

D. 0.406

E. 0.560

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347. In a 95 percent confidence interval, which of the following is the standard error of the difference of the two sample means between all comedy movie downloads and all drama movie downloads from the streaming service?

A film student was interested in whether there was a difference in the mean file sizes of all movie downloads from a popular streaming service for two genres—comedy and drama. The film student selected a random sample of comedy movies and a random sample of drama movies from the streaming service. The summary statistics are shown in the table.

	Number in Sample	Mean (in megabytes)	Standard Deviation (in megabytes)
Comedy	32	468	113
Drama	41	508	141

A. $\sqrt{\frac{113^2}{32} + \frac{141^2}{41}}$

B. $\sqrt{\frac{113^2}{32} - \frac{141^2}{41}}$

C. $\sqrt{\frac{113^2+141^2}{32+41}}$

D. $\sqrt{\frac{113^2-141^2}{32+41}}$

E. $\frac{113}{\sqrt{32}} + \frac{141}{\sqrt{41}}$

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348. Of the following, which is the least number of drivers in the community who should be surveyed such that the margin of error of the 95 percent confidence interval will be no greater than 5 percentage points?

Ashton will construct a 95 percent confidence interval to estimate the proportion of all drivers in his community who would indicate that they find it unacceptable that other drivers talk on a phone at any time while driving.

A. 30

B. 300

C. 400

D. 500

E. 600

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349. Which of the following must be true for an inference procedure that uses the same sample data?

An Internet service provider wants to know whether its customers would upgrade to a new Internet service. The Internet service provider collected data from a random sample of its customers. The customers in the sample indicated whether they were interested in the upgrade. Based on the sample data, a hypothesis test found convincing statistical evidence at the 5 percent level of significance that the proportion of customers who would upgrade to the new Internet service is less than 0.40.

A. A two-sided hypothesis test conducted at the 5% level of significance would find convincing statistical evidence that the proportion of customers who would upgrade to the new Internet service is different than 0.40.

B. A one-sided hypothesis test conducted at the 10% level of significance would find convincing statistical evidence that the proportion of customers who would upgrade to the new Internet service is less than 0.40.

C. A one-sided hypothesis test conducted at the 1% level of significance would find convincing statistical evidence that the proportion of customers who would upgrade to the new Internet service is less than 0.40.

D. A 95% confidence interval for the proportion of customers who would upgrade to the new Internet service would include 0.40.

E. A 90% confidence interval for the proportion of customers who would upgrade to the new Internet service would include 0.40.

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350. Which of the labeled points has a predicted lead level in the lab that is between 20 mcg and 40 mcg and has a positive residual?

The California condor is an endangered bird species. One reason the species is endangered is because of lead poisoning. It is costly and time-consuming to use lab testing to measure lead levels of condors. However, lab testing is not necessary if testing for lead in the wild is accurate. A biologist is interested in investigating the accuracy of lead level measurements taken in the wild. The biologist measured the lead level, in micrograms (mcg), for 14 California condors both in the wild and in the lab.

The scatterplot and least-squares regression line for predicting the lead level in the lab based on the lead level in the wild are shown. Five points are labeled.

Condor Lead Level Measurements

- There are 14 data points in the scatterplot, five of which are labeled A, B, C, D, and E.
- The data points begin in the lower left portion of the scatterplot and trend upward and to the right.
- The line of best fit

  begins at the point with coordinates 2 comma 7.5 and slants upwards and to the right. It passes through the point with coordinates 15 comma 20 and ends at the point with coordinates 47.5 comma 52.
- The data points labeled A, B, and C are above the line of best fit.
- The data points labeled D and E are below the line of best fit.
- The data represented by the five labeled points are as follows. Note that all values are approximate.
  - Point A: 10 comma 30.
  - Point B: 12 comma 38.
  - Point C: 16 comma 23.
  - Point D: 19 comma 22.
  - Point E: 47.5 comma 45.

A. A

B. B

C. C

D. D

E. E

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351. Which of the following statistics would not have a different numerical value for the weights in kilograms?

The weights of 100 cats at a veterinary clinic were recorded. Summary statistics for the weights of the 100 cats, in pounds, are shown in the table.

Summary Statistics for Weights of the Cats (in pounds)

Mean	Median	StDev	Min	Max	Q1	Q3
8.54	8.53	1.49	6.06	10.96	7.10	9.75

The weights of the 100 cats were converted from pounds to kilograms, and 1 pound is approximately equal to 0.45 kilogram.

A. The mean

B. The median

C. The standard deviation

D. The interquartile range

E. The standardized score (z-score) for a cat at the veterinary clinic

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352. Which of the following conclusions is supported by the interval?

An owner of a large number of car dealerships is concerned that salespeople's time using technology (cell phones and computers) for personal use during work hours is negatively affecting car sales. The owner collected data for 54 randomly selected salespeople. The scatterplot for the linear regression analysis predicting mean unit sales per week from mean time using technology for personal use during work hours, in minutes per day, is shown.

Scatterplot of Data Collected for Each Salesperson

- There are 54 data points in the scatterplot.
- The data points are scattered throughout the plane, trending slightly downward and to the right.
- The line of best fit:
  - Begins at the point with coordinates 10 comma 5.8, and slants gradually downward and to the right
  - Passes through the points with coordinates 30 comma 5, and 62 comma 4
  - Ends at the point with coordinates 90 comma 3
- Note that all values are approximate.

Assuming the conditions for inference have been met, a 95 percent confidence interval for the slope of the population regression line was found to be $(-0.0525, -0.0124)$.

A. The mean unit sales per week for all salespeople at the car dealerships is between 0.0124 and 0.0525.

B. Because the interval does not contain 0, there is not convincing statistical evidence of a linear relationship between mean time using technology for personal use and mean unit sales per week for all salespeople at the dealerships.

C. Because the interval does not contain 0, there is convincing statistical evidence that there is no linear relationship between mean time using technology for personal use and mean unit sales per week for all salespeople at the dealerships.

D. Because the interval does not contain -1, there is convincing statistical evidence that there is no linear relationship between mean time using technology for personal use and mean unit sales per week for all salespeople at the dealerships.

E. Because the values in the interval are negative, there is convincing statistical evidence of a decrease, on average, in mean unit sales per week for each 1-minute increase in mean time using technology for personal use for all salespeople at the dealerships.

353.

A. Describe the relationship between the amount of flour and the number of cookies produced, in context.

B. The least-squares regression equation for predicting the number of cookies produced (y) based on the amount of flour (x) is $\hat{y} = 3.61 + 19.57x$.

Identify and interpret the y-intercept of the least-squares regression equation, in context.

C. The residual plot for the least-squares regression model is shown.

- There are 22 data points in the scatterplot.
- The data points are scattered throughout the scatterplot.
- The leftmost data point is at coordinates 1 comma negative 3
- The uppermost data point is at coordinates 3.1 comma 5.9
- The rightmost data points are at coordinates 4 comma 0 and 4 comma 2.1
- The lowermost data point, which is circled, is at coordinates 2.4 comma negative 10.58.
- Note that all values are approximate.

Consider the circled point on the residual plot with a residual of -10.58 cookies produced.

i. What is the predicted number of cookies produced by the recipe represented by the circled point? Show how you obtained your answer.

ii. What was the actual number of cookies produced by the recipe represented by the circled point? Show how you obtained your answer.

A baker is investigating the number of cookies that can be produced based on the amount of flour that is used in a recipe. The baker selected a random sample of 22 different chocolate chip cookie recipes that use flour. The cookies produced are each 2 inches in diameter. The amount of flour, in cups, and the number of cookies produced were recorded for each recipe. The scatterplot of the data for the 22 chocolate chip cookie recipes is shown.

Chocolate Chip Cookie Recipes

- There are 22 data points in the scatterplot.
- Two of the data points are in the lower left portion of the scatterplot at the points with coordinates:
  - 1 comma 20
  - 1.5 comma 37
- The remaining data points begin in middle left portion of the scatterplot at the point with coordinates 2 comma 47, and trend upward and to the right in a tight cluster until they end in the top right part of the scatterplot at the point with coordinates 4 comma 85.
- Note that all values are approximate.

354.

A. Identify the population of interest and explain why it is reasonable to generalize the researchers' conclusion to the population of interest, in context.

B. Explain, in context, why this study cannot be double-blind.

C. Another researcher, Dr. Adams, believes that adults over 65 years old with a certain genetic trait may respond differently to type of exercise (i.e., aerobic or strength training) than adults over 65 years old without the trait. Dr. Adams proposes a new study using a design with one block consisting of adults with the trait and a second block consisting of adults without the trait. Within each block, the participants will be randomly assigned to either the aerobic only exercise group or the strength-training-only exercise group. The total amount of sleep over seven days will be recorded for each adult.

If Dr. Adams' belief is correct, explain why Dr. Adams' proposed study is likely better able than the original study to detect a difference in the population means for the amount of total sleep between the two types of exercise.

A random sample of adults over 65 years-old from a certain community participated in a study investigating the relationship between type of exercise and amount of sleep. Participants in the study were randomly assigned to one of two exercise groups: aerobic exercise only or strength training only. Over seven days, the total amount of sleep, in hours, was recorded for each participant. The researchers found that the conditions for a two-sample *t*-test for the difference in population means were met, and they carried out the hypothesis test. The researchers concluded that the mean amount of total sleep was significantly greater for participants assigned to the aerobic exercises than for participants assigned to the strength-training exercises.

355.

A. Elida is an outstanding snowboarder who will participate in many snowboarding competitions in the upcoming season. In each competition, snowboarders with the three top scores win a medal. Elida has a 55 percent chance of winning a medal in each of her future competitions.

i. Elida will participate in one competition next week. What is the probability that Elida will not win a medal in that competition?

ii. Assuming the results of her competitions are independent and that she will compete in 23 competitions this season, what is the expected number of medals that Elida will win? Show your work.

B. In the preseason, Elida will compete in 10 competitions. Assume the results of her competitions are independent and her chance of winning a medal in each of the preseason competitions is 55 percent. Let X represent the number of medals she will win out of the 10 competitions.

i. Calculate $P(X \geq 8)$. Show your work.

ii. Interpret the probability calculated in part B (i).

C. Carmen is a snowboarder who will participate in a world class competition. The snowboarding event has a qualification round and a final round. The twelve snowboarders with the top scores in the qualification round advance to the final round. The three snowboarders with the top scores in the final round win a medal. Suppose that Carmen will advance to the final round. The table provides information about Carmen's world class competition probabilities.

Carmen's World Class Competition Probabilities

	Wins a Medal in the Final Round	Does Not Win a Medal in the Final Round	Total
Earns a score among the top five in the qualification round	0.624	0.156	0.78
Does not earn a score among the top five in the qualification round	0.088	0.132	0.22
Total	0.712	0.288	1.0

Consider the information provided in the table.

i. Are the events "earns a score among the top five in the qualification round" and "wins a medal in the final round" mutually exclusive? Justify your answer.

ii. Are the events "earns a score among the top five in the qualification round" and "wins a medal in the final round" independent? Justify your answer using probabilities computed from the table.

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356. Is there convincing statistical evidence, at a 5 percent significance level, that there is a difference in the distributions of preferred discount method between age groups? Justify your answer with the appropriate inference procedure.

A grocery store chain would like to stop using paper coupons to provide discounts on items. Instead, customers would use a cell-phone application (app) to claim a discount.

Carlos is a manager at one of the chain's largest grocery stores. He wants to determine whether there is a difference in preferred discount method among customers of different ages. To investigate, he takes separate random samples of 100 customers from each of two age groups:

- 50 years old and younger
- Older than 50 years

The grocery store has more than 2,000 customers in each age group. Carlos asks each sampled customer whether they would prefer to use the app to claim a discount, to use paper coupons to claim a discount, or to receive no discount at all. Carlos' results are recorded in the following table.

Preferred Discount Method by Age Group

Age-Group	Use the App	Use Paper Coupons	No Discount	Total
50 years old and younger	64	22	14	100
Older than 50 years	50	44	6	100
Total	114	66	20	200

357.

A. A candy apple produced by the bakery will be selected at random. What is the probability that the candy apple weighs more than 170 grams? Show your work.

B. Marissa wants to use a new variety of apple that is larger than the current variety of apple so that she can charge more money for the candy apples. Marissa claims that the population mean weight of the candy apples produced with the new variety of apples will be greater than 160 grams. A random sample of *n* candy apples produced using the new variety had a mean weight of 170 grams. To test whether the new variety produces candy apples with a mean weight that is greater than 160 grams, a simulation was conducted with 1,000 samples of size *n* from a population with a mean of 160 grams and a standard deviation of 18 grams. For the simulation, the mean of each sample was calculated. The results of the simulation are shown in the graph.

Simulation of 1,000 Samples

- There are 16 bars in the histogram.
- The data represented by the bars are as follows:
  - 140 grams up to but not including 142.5 grams, 2.
  - 142.5 grams up to but not including 145 grams, 4.
  - 145 grams up to but not including 147.5 grams, 8.
  - 147.5 grams up to but not including 150 grams, 20.
  - 150 grams up to but not including 152.5 grams, 48.
  - 152.5 grams up to but not including 155 grams, 88.
  - 155 grams up to but not including 157.5 grams, 148.
  - 157.5 grams up to but not including 160 grams, 186.
  - 160 grams up to but not including 162.5 grams, 174.
  - 162.5 grams up to but not including 165 grams, 148.
  - 165 grams up to but not including 167.5 grams, 103.
  - 167.5 grams up to but not including 170 grams, 46.
  - 170 grams up to but not including 172.5 grams, 15.
  - 172.5 grams up to but not including 175 grams, 6.
  - 175 grams up to but not including 177.5 grams, 2.
  - 177.5 grams up to but not including 180 grams, 2.

State the hypotheses that should be used to test Marissa's claim.

C. Based on the simulation, is Marissa's claim supported at the 5 percent level of significance? Justify your answer.

D. Consider your response to part C.

i. Identify and describe the type of error—Type I or Type II—that would be possible. State your answer in the context of the problem.

ii. Recall that Marissa wants to use a new variety of apple that is larger than the current variety of apple so that she can charge more money for the candy apples. Based on the response given in part D (i), what is a consequence of the error for Marissa's bakery?

Marissa owns a bakery that produces 250 candy apples every day using a certain variety of apples. The distribution of the weight of the candy apples produced by the bakery is approximately normal with a mean of 160 grams and a standard deviation of 18 grams.

358.

A. Consider the dotplot.

i. Approximately, what is the numeric value of point A?

ii. Interpret the value of point A, in context.

B. When the oxygen saturation levels measured by pulse oximeter and by blood test are the same, then the mean difference will be zero. After looking at the dotplot, the doctor is more concerned that the sample data indicate there is a nonzero mean difference in the oxygen saturation between the pulse oximeter and the blood test for the population. The differences of oxygen saturation (pulse oximeter minus blood test) measurements for the 25 individuals in the sample have a mean of -1.28 points and a standard deviation of 0.724 points.

i. Assume the conditions for inference have been met. Construct a 95 percent confidence interval for the population mean difference for oxygen saturation measured by pulse oximeter and oxygen saturation measured by blood test (pulse oximeter minus blood test). Show your work.

ii. Based on the interval in part B (i), is there support for the doctor's concern? Justify your answer.

C. The doctor wants to further explore how these two methods differ using a Tukey mean-difference plot. A Tukey mean-difference plot is a scatterplot that shows how the difference in two paired measurements is related to the mean of the two measurements.

To create this plot, for each of the 25 individuals in the sample, the difference of the two oxygen saturation measurements (pulse oximeter minus blood test) and the mean of the two oxygen saturation measurements are plotted. The Tukey mean-difference plot for the doctor's sample is shown with two points circled, B and C.

The following shows the computation of the mean and the difference for the individual represented by point B.

$$\text{Mean} = \frac{78 + 79.8}{2} = 78.7 \text{ points and Difference} = 78 - 79.8 = -1.4 \text{ points.}$$

The computations of the mean and difference for each of the other 24 individuals have been calculated and are plotted in the following Tukey mean-difference plot.

Tukey Mean-Difference Plot

- There are 25 data points in the scatterplot.
- The data points:
  - Begin in the bottom left part of the scatterplot at the point with coordinates 71.5 comma negative 2.8
  - Trend upward and to the right
  - End in the top right part of the scatterplot at the point with coordinates 94 comma 0
- Two data points are circled and labeled B and C.
  - The coordinates of point B are 78 comma negative 1.4
  - The coordinates of point C are 94 comma 0
- Note that all values are approximate.

Consider the Tukey mean-difference plot for the doctor's data.

i. What is the approximate mean oxygen saturation and the approximate difference in oxygen saturation for point C?

ii. Compare the mean oxygen saturation for points B and C.

iii. For each point B and C, does the pulse oximeter result in an oxygen saturation estimate that is higher than, lower than, or the same as the estimate for the blood test? Explain your answer in context.

D. In the doctor's Tukey mean-difference plot, there is a positive relationship between the differences in oxygen saturation and the mean oxygen saturation.

i. Explain what the positive relationship in the plot indicates about the accuracy of the pulse oximeter method of measuring oxygen saturation relative to the blood test method.

ii. The doctor will use the pulse oximeter to measure oxygen saturation for new patients because it is more convenient and less invasive. Suppose one of the doctor's new patients has a pulse oximeter measurement of 73 points. Based on your answer to part D (i), should the doctor trust the measurement? Justify your answer.

The amount of oxygen in blood, oxygen saturation, is an important factor in determining a person's health. The oxygen saturation is measured as a score from 0 to 100 points. A doctor is comparing two methods for measuring a person's oxygen saturation.

- Standard method: A blood test is used to measure oxygen saturation.
- Alternative method: A pulse oximeter, a device that clips to the index finger, uses light to measure oxygen saturation.

While the blood test is very accurate, using a pulse oximeter is a more convenient and less invasive method for measuring oxygen saturation. The doctor is concerned, *however*, that the pulse oximeter measurement may not equal the blood test measurement. For the investigation, the doctor randomly sampled 25 individuals from a large population and measured each person's oxygen saturation using both methods. For each individual, the doctor computed the difference in oxygen saturation measurements (pulse oximeter minus blood test). The differences in oxygen saturation measurements are displayed in the dotplot shown, with a particular point circled and labeled A to represent a person in the study.

Difference in Oxygen Saturation Measurements Using Two Methods for 25 Sampled Individuals

- One dot is circled and labeled A, which is indicated directly to the left of negative 2.
- The remainder of the dots are as follows.
  - negative 2.8: 1 dot
  - negative 2.5: 1 dot
  - negative 2.4: 1 dot
  - negative 1.8: 1 dot
  - negative 1.7: 2 dots
  - negative 1.6: 2 dots
  - negative 1.5: 1 dot
  - negative 1.4: 1 dot
  - negative 1.3: 2 dots
  - negative 1.2: 1 dot
  - negative 1.1: 2 dots
  - negative 0.9: 3 dots
  - negative 0.7: 2 dots
  - negative 0.5: 1 dot
  - negative 0.2: 1 dot
  - negative 0.1: 1 dot
  - 0: 1 dot